speculation and we have no ability to verify it, since the phenomenon is most pronounced on the large proprietary models where sometimes not only the model weights, but also the tokenizers are hidden. We still think the discussion is worth it.

It has been observed for some time that LLMs can interpret sequences with fillers, like `a!b!r!a!c!a!d!a!`. While the exact reasons behind these phenomenon are in our opinion not quite explored, it is possible that it is explained by the process depicted on Figure 3: there are some mechanisms [Wan+22] that identify and remove the filler tokens, guess relevant substitution cipher ROT_n [3] and map the result to the same concept [Elh+22; Tem+24].

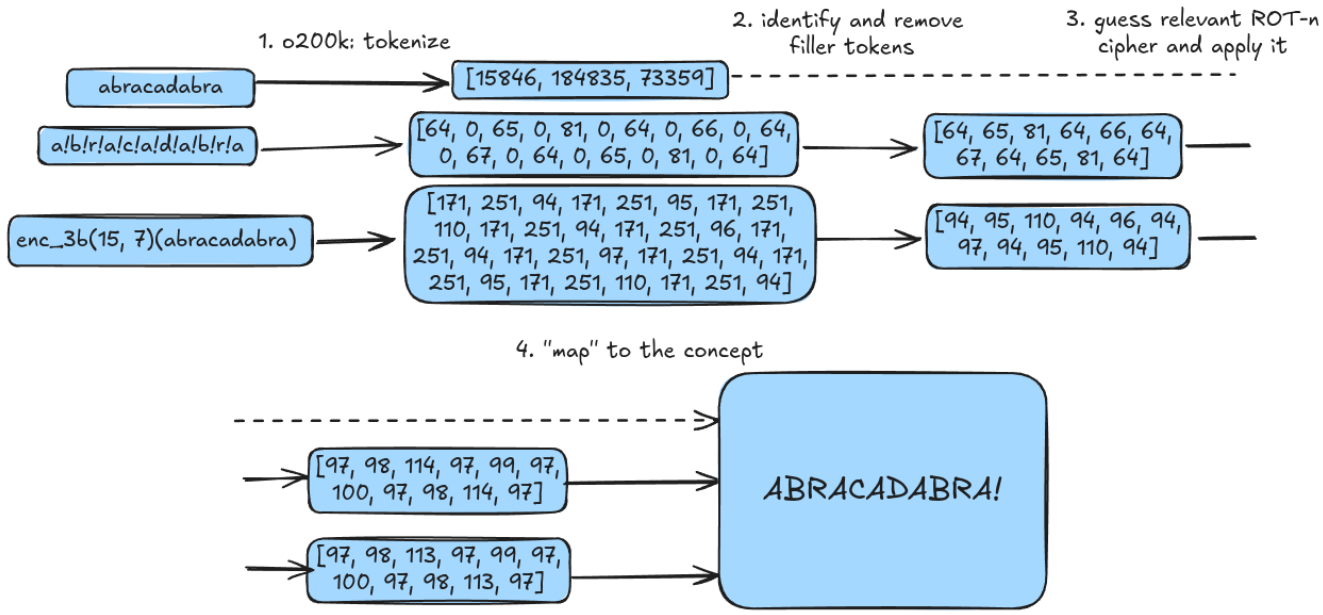

Figure 3: Our initial hypothesis on the underlying reason. The third example is given here

This hypothesis would in particular explain, how the models learn to understand the text written in [Goo24].

Reasoning models

Here we try summarize what we observed in reasoning models. They were extremely good at answering such requests. In particular, Deepseek-R1 chain-of-thought contained suggestions to use substitution cipher and some time after it successfully decoded the message. We did not conduct a systematic study.

4 Related Work

In this section we discuss the most relevant references on Unicode abilities of LLMs and generalization of adversarial training.

The most relevant references seem to be the work noting the existence of classical software vulnerabilities connected to the text wrapped inside of Unicode Tags (which therefore appears

[3] which replaces each symbol l by the symbol that is n-th after it in ASCII table $\mathrm{ROT}_n(l) = \mathrm{chr}(\mathrm{ord}(l) + n)$

invisible to human eye) in [BA23] and the observation that LLMs can interpret such symbols in [Goo24]. Another relevant observation is about models being not able to read texts aligned from top to bottom [Li+24]: we actually found anecdotal evidence that some models could understand similarly formatted texts.

There have been discovered many failure modes of adversarial training: extreme sensitivity to the tense of the instruction [AF24], misgeneralization to function calling [Wu+24], vulnerability to various string transformations [Hua24], different variants of optimization of adversarial prompts by automated methods [Zou+23]. The fact that alignment mechanisms didn't turn out to be that fragile against our attack is encouraging.

Our work is different because it provides a large class of abilities/vulnerabilities, which, while illegible to untrained human eye, are nonetheless clearly interpretable.

5 Discussion, Limitations, and Future Work

We think this work raises several questions worth of discussion, some of them quite speculative, but nonetheless important:

- It seems to be an experimental confirmation of the invalidity of ML approach to LLM Censorship [Glu+23]
- Non-LLM input/output judges are likely to become obsolete upon the advent of LLMs with such abilities, at least for the purposes of safety and security, since models now can understand too wide a class of encodings (and possibly communicate using them). Note that the situation is quite different from communicating in Base64 etc, since for those cases we in principle could enumerate major known to humanity encodings and check that the model doesn't say anything bad.
- At the same time, LLMs' ability to output texts that are illegible to other LLM judges by construction forces us to use the same model to judge itself. This situations have been argued to possess some inherent self-preference bias [PBF24].
- If the other LLM, in particular, is a weaker versions of the same LLM during alignment in the spirit of [Bur+23], the problem might get worse since the larger model might learn to switch to the language that the overseer doesn't understand; see also the discussion of scalable oversight issues at B.4 of [Wan+24]. It might be bad as long as LLMs do not converge in their abilities [Huh+24]
- Warrants careful assessment of similar fallacies during evaluations of scheming [Ant24c] and other ASL-4 type behaviours [Jon+25].
- It shows that two directions of research probably deserve more attention: interpretability of LLMs [Gup+24] and non-tokenization approaches [Pag+24].

This work would benefit from a wider variety of encoding classes, more extensive adversarial evaluations and, especially, mechanistic interpretability of the observed phenomena. We hope that these limitations will be addressed in the future.

We decided to publish the results without first disclosing them to the model providers, since 1) we don't expect the underlying cause to be fixed easily (for example, it's hardly a solution to block most of the Unicode table, and adversarial training on the provided examples will probably be superficial only) 2) we saw the significant growth of these abilities in the Claude family models, while finding early indications of growth in almost every major large enough LLM we tested 3) we are not sure where the boundary of abilities/vulnerabilities lies in this case.

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